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# Notes: a tutorial on DMFT and the cavity method for a really simple example

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Emmy Blumenthal

Email: eblu (at) princeton (dot) edu

Princeton University Department of Physics

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**Objective.** This note aims to explain what dynamical mean-field theory (DMFT) is and how it can be derived using the cavity method. I do this for a very simple example of a linear system so that we can compare with the exact solution. At the end, I show a nice relation to the computation of the spectral densities of random matrices. Some prior knowledge of random matrix theory (RMT) and Gaussian processes is helpful but not strictly necessary. Most of my understanding of these topics comes from Refs. [1, 2, 3, 4, 5, 6, 7], many of which concern ecological models.

If you find any mistakes, typos, or have suggestions for improvement, please let me know.

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## 1 Linear dynamics

### 1.1 Setup

We'll study the following model of linear dynamics of  $N$  constituents:

$$\frac{dx_i}{dt} = \sum_{j=1}^N A_{ij} x_j(t), \quad i = 1, \dots, N, \quad (1)$$

where  $A_{ij}$  is a skew-symmetric ( $A_{ij} = -A_{ji}$ ) random matrix and  $x_i(0)$  is a random vector. We will assume  $A_{ij}$  and  $x_i(0)$  are normally distributed with statistics:

$$\begin{aligned} \langle A_{ij} \rangle &= 0, & \langle A_{ij} A_{kl} \rangle &= \frac{\sigma^2}{N} (\delta_{ij} \delta_{kl} - \delta_{il} \delta_{jk} (1 - \delta_{ij})) \\ \langle x_i(0) \rangle &= 0, & \langle x_i(0) x_j(0) \rangle &= \sigma_0^2 \delta_{ij} \end{aligned} \quad (2)$$

Everything we do here will also work for the more general case  $\langle A_{ij} A_{kl} \rangle = \frac{\sigma^2}{N} (\delta_{ij} \delta_{kl} + \gamma \delta_{il} \delta_{jk} (1 - \delta_{ij}))$  with  $\gamma \in [-1, 1]$ , but we focus on the case  $\gamma = -1$  for simplicity and because it has persistent, bounded dynamics whereas the other cases either decay or grow exponentially.

## 1.2 Useful properties of random matrices

These sorts of matrices that have  $\langle A_{ij} \rangle \sim O(1/N)$  and  $\text{var}[A_{ij}] \sim O(1/N)$  have several useful properties as  $N \rightarrow \infty$  [7]. This scaling is often a good choice in many applications because it leads to a well-defined thermodynamic limit. For example, in the above, it ensures that all derivatives are  $O(1)$  when  $x_i(t)$  is  $O(1)$ . We define the spectral density of a random matrix  $A$  as:

$$\rho_A(x) = \frac{1}{N} \sum_{i=1}^N \delta(x - \lambda_i), \quad (3)$$

where  $\lambda_i$  are the eigenvalues of  $A_{ij}$ . One of the most important properties of these sorts of matrices is that all the eigenvalues are  $O(1)$  and the spectral density converges to a deterministic (usually) well-behaved function as  $N \rightarrow \infty$ . More precisely,  $\text{var}[\rho_A(x)] \sim 1/N$  as  $N \rightarrow \infty$ . This additionally means that for an  $O(1)$  analytic function  $f$ , we have:

$$\frac{1}{N} \text{Tr} f(A) \rightarrow \int dx \rho_A(x) f(x) = O(1) + \text{fluctuations of order } O(1/\sqrt{N}). \quad (4)$$

This is a consequence of the central limit theorem and the law of large numbers. In particular, it means that we can always replace  $\frac{1}{N} \text{Tr} f(A)$  with its expected value under resampling the matrix  $A_{ij}$  in the limit  $N \rightarrow \infty$ .

For our matrix  $A_{ij}$ , the spectral density is given by:

$$\rho_A(x + iy) = \begin{cases} \delta(x) \cdot \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - y^2}, & \text{if } -2\sigma \leq y \leq 2\sigma, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The  $\delta(x)$  indicates that the spectral density exists only on the imaginary axis. I recommend reading Ref. [7] for a more detailed discussion of these properties.

## 1.3 Dynamical mean-field theory

### 1.3.1 Solving exactly.

**What is DMFT?** Dynamical mean-field theory (DMFT) is a method for studying the dynamics of systems with many constituents. DMFT is a description of the effective dynamics of a single constituent in the system which is coupled to a mean-field that is determined by the other constituents. DMFT in this sort of system is a little different from the usual mean-field theory you first learn in undergraduate statistical mechanics. In this sort of system, we are not going to treat every degree of freedom the same way: they are composed of heterogeneous constituents. Therefore, we write down the dynamics for a single randomly chosen constituent and integrate out the other constituents. We will call disorder due to randomly selecting the constituent of interest *shuffling* randomness which is different from the *quenched* disorder that comes from sampling the matrix  $A_{ij}$ . We will use the convention of calling the randomly selected constituent  $i = \alpha$ . The shuffled average of some quantity  $X$  is defined as:

$$\langle X \rangle_{\text{sh}} = \frac{1}{N} \sum_{\alpha=1}^N (X \text{ with } \alpha \text{ being the chosen constituent}) \quad (6)$$

We will write averages over the quenched disorder as:

$$\langle X \rangle_{\mathcal{Q}} = \int \prod_{i,j=1}^N dA_{ij} \prod_{i=1}^N dx_i(0) p(\{A_{ij}\}) p(\{x_i(0)\}) X \quad (7)$$

DMFT results in a single equation of motion (often a stochastic differential equation) for the randomly chosen constituent  $\alpha$ . Resampling the stochastic dynamics is meant to capture shuffling randomness. When

you can replace a quantity with its quenched average, one says that a particular quantity is self-averaging. Sometimes it is said that an entire system/model is self-averaging if all the shuffled averages of all quantities of interest can be replaced with their quenched averages. This is related to replica symmetry, but in the context of spin glasses, this has a slightly more specific meaning, so the use of the term self-averaging is more widely applicable. Note: there are many ways to discuss, define, and interpret DMFT. This is just one way to do it.

Because this system is linear, almost every detail here is exactly solvable. This means we will be able to make many arguments about properties of the system starting from the microscopic details. In other nonlinear contexts, these arguments are not so straightforward and are instead taken as assumptions. This is to say that when you encounter “real-life” DMFT, you will not be able to do things as nicely as we do here.

**Integrating out  $N - 1$  degrees of freedom.** We will now do something you can only do in a linear system: we will explicitly integrate out the  $N - 1$  degrees of freedom. After selecting the randomly chosen constituent  $\alpha$ , we can write down the dynamics of the system as:

$$\begin{aligned}\frac{dx_i}{dt} &= \sum_{j(\neq\alpha)} A_{ij}x_j(t) + A_{i\alpha}x_\alpha(t), \quad (i \neq \alpha), \\ \frac{dx_\alpha}{dt} &= \sum_{i(\neq\alpha)} A_{\alpha i}x_i(t) + A_{\alpha\alpha}x_\alpha(t)\end{aligned}\tag{8}$$

Because this is a linear system, we can exactly integrate out the  $N - 1$  degrees of freedom:

$$x_i(t) = \sum_{j(\neq\alpha)} [e^{A_{\setminus\alpha}t}]_{ij}x_j(0) + \int_0^t dt' \sum_{j(\neq\alpha)} [e^{A_{\setminus\alpha}(t-t')}]_{ij}A_{j\alpha}x_\alpha(t')\tag{9}$$

where  $A_{\setminus\alpha}$  is the matrix  $A_{ij}$  with the  $\alpha$ th row and column removed. We can then substitute this into the equation for  $x_\alpha(t)$ :

$$\frac{dx_\alpha}{dt} = \Phi_\alpha(t) - \sigma^2 \int_0^t dt' \nu_\alpha(t-t')x_\alpha(t')\tag{10}$$

where we have defined:

$$\begin{aligned}\Phi_\alpha(t) &\equiv \sum_{i,j(\neq\alpha)} A_{\alpha i}[e^{A_{\setminus\alpha}t}]_{ij}x_j(0) \\ \nu_\alpha(t) &\equiv -\frac{1}{\sigma^2} \sum_{i,j(\neq\alpha)} A_{\alpha i}[e^{A_{\setminus\alpha}t}]_{ij}A_{j\alpha}\end{aligned}\tag{11}$$

Note that we have dropped the  $A_{\alpha\alpha}x_\alpha(t)$  term because it is  $O(1/\sqrt{N})$  and therefore negligible in the limit  $N \rightarrow \infty$ .

**Averages and statistical analysis.** We begin by looking at  $\Phi_\alpha(t)$ . We assert (perhaps handwavingly) that it is a Gaussian process by the central limit theorem because it is a sum of many independent random variables (the  $A_{ij}$ ’s).<sup>1</sup> There are more rigorous ways to show this. For fixed  $\alpha$ , the origin of the Gaussian process-ness and thus fluctuations in  $\Phi_\alpha$  is from the quenched disorder (i.e., the  $A_{ij}$ ’s) at fixed shuffling

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<sup>1</sup>In the case where  $A_{ij}$  are Gaussian, this is exactly true, and it is not necessary to invoke the central limit theorem.

randomness. We will then proceed to compute its statistics using a quenched average:

$$\begin{aligned}
\langle \Phi_\alpha(t) \rangle_{\mathcal{Q}} &= \sum_{i,j(\neq \alpha)} \langle A_{\alpha i} \rangle \langle [e^{A_{\setminus \alpha} t}]_{ij} \rangle \langle x_j(0) \rangle = 0 \\
\langle \Phi_\alpha(t) \Phi_\alpha(s) \rangle_{\mathcal{Q}} &= \sum_{i,j,k,l(\neq \alpha)} \langle A_{\alpha i} A_{\alpha k} \rangle \langle [e^{A_{\setminus \alpha} t}]_{ij} [e^{A_{\setminus \alpha} s}]_{kl} \rangle \langle x_j(0) x_l(0) \rangle \\
&= \sigma^2 \sigma_0^2 \frac{1}{N} \sum_{i,j(\neq \alpha)} \langle [e^{A_{\setminus \alpha} t}]_{ij} [e^{A_{\setminus \alpha} s}]_{ij} \rangle = \sigma^2 \sigma_0^2 \frac{1}{N} \text{Tr} \left[ e^{A_{\setminus \alpha} t + A_{\setminus \alpha}^T s} \right] \\
&= \sigma^2 \sigma_0^2 \int d^2 \lambda \rho_A(\lambda) e^{\lambda t + \lambda^* s} \\
&= \frac{\sigma_0^2}{2\pi} \int_{-2\sigma}^{2\sigma} dy \sqrt{4\sigma^2 - y^2} e^{iy(t-s)} = \sigma_0^2 \sigma^2 \frac{J_1(2\sigma|t-s|)}{\sigma|t-s|}
\end{aligned} \tag{12}$$

Note that we were able to factor the averages of  $A_{\alpha i}$  and  $e^{A_{\setminus \alpha} t}$  as the two are definitionally independent. Now, what if we fixed  $A_{ij}$  (making  $\Phi_\alpha(t)$  no longer a random variable) and we instead average over the shuffling randomness? Well that's hard to analyze because we can't easily take averages over the shuffling randomness. However, we might wonder what the statistics of the shuffled average is under quenched disorder:

$$\begin{aligned}
\langle \langle \Phi_\alpha(t) \rangle_{\text{sh}} \rangle_{\mathcal{Q}} &= \frac{1}{N} \sum_{\alpha=1}^N \langle \Phi_\alpha(t) \rangle_{\mathcal{Q}} = 0, \\
\langle [\langle \Phi_\alpha(t) \rangle_{\text{sh}}]^2 \rangle_{\mathcal{Q}} &= \frac{1}{N^2} \sum_{\alpha,\beta=1}^N \langle \Phi_\alpha(t) \Phi_\beta(t) \rangle_{\mathcal{Q}} \\
&= \frac{1}{N^2} \sum_{\alpha,\beta} \sum_{i,j(\neq \alpha)} \sum_{k,l(\neq \beta)} \langle A_{\alpha i} A_{\beta k} \rangle \langle [e^{A_{\setminus \alpha} t}]_{ij} [e^{A_{\setminus \beta} t}]_{kl} \rangle \langle x_j(0) x_l(0) \rangle \\
&= \frac{\sigma^2 \sigma_0^2}{N^3} \sum_{\alpha,\beta} \sum_{i,j(\neq \alpha)} \sum_{k,l(\neq \beta)} (\delta_{\alpha\beta} \delta_{ik} - \delta_{\alpha k} \delta_{\beta i} (1 - \delta_{\alpha\beta})) \langle [e^{A_{\setminus \alpha} t}]_{ij} [e^{A_{\setminus \beta} t}]_{kl} \rangle \delta_{jl} \\
&= \frac{\sigma^2 \sigma_0^2}{N^3} \left( \sum_{\alpha} \sum_{i,j(\neq \alpha)} \langle [e^{A_{\setminus \alpha} t}]_{ij} [e^{A_{\setminus \alpha} t}]_{ij} \rangle - \sum_{\alpha,\beta(\alpha \neq \beta)} \sum_{j(\neq \alpha,\beta)} \langle [e^{A_{\setminus \alpha} t}]_{\beta j} [e^{A_{\setminus \beta} t}]_{\alpha j} \rangle \right) \\
&= \frac{\sigma^2 \sigma_0^2}{N^3} \left( \sum_{\alpha} \text{Tr} \left[ e^{A_{\setminus \alpha} t} e^{A_{\setminus \alpha}^T t} \right] - \sum_{\alpha,\beta(\alpha \neq \beta)} \sum_{j(\neq \alpha,\beta)} \langle [e^{A_{\setminus \alpha} t}]_{\beta j} [e^{A_{\setminus \beta} t}]_{\alpha j} \rangle \right) \\
&\sim \frac{1}{N^3} \cdot N^2 \sim O\left(\frac{1}{N}\right).
\end{aligned} \tag{13}$$

In the above, we have used the fact that the trace of a random matrix with the appropriate scaling is  $O(N)$  because it is a sum of  $N$   $O(1)$  eigenvalues. The first term in the second-to-last line we could write as an explicit trace but the second term is a bit more notationally complicated just because the trace is over the  $(\alpha, \beta)$  indices; nonetheless, the final result is  $O(1/N)$ . This means that the fluctuations in the shuffled average between resampling the matrix  $A_{ij}$  become negligible in the limit  $N \rightarrow \infty$  (i.e.,  $\text{var}[\langle \Phi_\alpha(t) \rangle_{\text{sh}}]_{\mathcal{Q}} \sim O(1/N)$ ). The same applies to  $\langle \langle \Phi_\alpha(t) \Phi_\alpha(s) \rangle_{\text{sh}} \rangle_{\mathcal{Q}}$ , but it's even more messy (but not conceptually different). We can then replace the shuffled average with the quenched average:

$$\begin{aligned}
\langle \Phi_\alpha(t) \rangle_{\text{sh}} &\rightarrow \langle \Phi_\alpha(t) \rangle_{\mathcal{Q}} \\
\langle \Phi_\alpha(t) \Phi_\alpha(s) \rangle_{\text{sh}} &\rightarrow \langle \Phi_\alpha(t) \Phi_\alpha(s) \rangle_{\mathcal{Q}}
\end{aligned} \quad \text{as } N \rightarrow \infty. \tag{14}$$

In general, we can do this for the moment generating function of  $\Phi_\alpha(t)$  which means that we can treat  $\Phi_\alpha(t)$  as a Gaussian process at fixed  $A_{ij}$  where the fluctuating Gaussian process corresponds to shuffling

randomness. This is the essence of replica symmetry. In most systems, you will not be able to derive replica symmetry in this way, but you will have to assume it.

We proceed to look at  $\nu_\alpha$  now. For the same reasons as  $\Phi$  above, we can replace the quenched average with the shuffled average, so we compute:

$$\begin{aligned}
\langle \nu_\alpha(t) \rangle_{\mathcal{Q}} &= -\frac{1}{\sigma^2} \sum_{i,j(\neq \alpha)} \langle A_{\alpha i} A_{j\alpha} \rangle \langle [e^{A_{\alpha} t}]_{ij} \rangle = \frac{1}{N} \sum_{i,j(\neq \alpha)} \delta_{ij} \langle [e^{A_{\alpha} t}]_{ij} \rangle = \frac{1}{N} \text{Tr} [e^{A_{\alpha} t}] \\
&= \int d^2 \lambda \rho_A(\lambda) e^{\lambda t} = \frac{1}{2\pi\sigma^2} \int_{-2\sigma}^{2\sigma} dy \sqrt{4\sigma^2 - y^2} e^{iyt} = \frac{J_1(2\sigma t)}{\sigma t} \\
\langle \nu_\alpha(t) \nu_\alpha(s) \rangle_{\mathcal{Q}} &= \frac{1}{\sigma^4} \sum_{i,j,k,l(\neq \alpha)} \langle A_{\alpha i} A_{j\alpha} A_{\alpha k} A_{l\alpha} \rangle \langle [e^{A_{\alpha} t}]_{ij} [e^{A_{\alpha} s}]_{kl} \rangle \\
&= \frac{1}{N^2} \sum_{i,j,k,l(\neq \alpha)} (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \langle [e^{A_{\alpha} t}]_{ij} [e^{A_{\alpha} s}]_{kl} \rangle \\
&= \left( \frac{1}{N} \text{Tr} [e^{A_{\alpha} t}] \right) \left( \frac{1}{N} \text{Tr} [e^{A_{\alpha} s}] \right) + \frac{1}{N^2} \text{Tr} [e^{A_{\alpha}(t-s)}] + \frac{1}{N^2} \text{Tr} [e^{A_{\alpha}(t+s)}] \\
&= \langle \nu_\alpha(t) \rangle_{\mathcal{Q}} \langle \nu_\alpha(s) \rangle_{\mathcal{Q}} + O\left(\frac{1}{N}\right) \implies \text{cov}[\nu_\alpha(t), \nu_\alpha(s)]_{\mathcal{Q}} = O\left(\frac{1}{N}\right).
\end{aligned} \tag{15}$$

It appears that the fluctuations (under all sources of disorder) in  $\nu_\alpha(t)$  are negligible in the limit  $N \rightarrow \infty$ , so we can replace it with its quenched average:

$$\nu_\alpha(t) \rightarrow \langle \nu_\alpha(t) \rangle_{\mathcal{Q}} = \frac{J_1(2\sigma t)}{\sigma t}. \tag{16}$$

**The effective dynamics.** Because we argued that  $\Phi_\alpha(t)$  is a Gaussian process at fixed  $A_{ij}$  under shuffling randomness, we can drop the  $\alpha$  subscript and write the effective dynamics of a randomly chosen constituent as:

$$\begin{aligned}
\frac{dx}{dt} &= \Phi(t) - \sigma^2 \int_0^t dt' \nu(t-t') x(t') \\
\Phi(t) &\sim \text{Gaussian Proc.}, \quad \langle \Phi(t) \rangle_{\text{sh}} = 0, \quad \langle \Phi(t) \Phi(s) \rangle_{\text{sh}} = \sigma^2 \sigma_0^2 \frac{J_1(2\sigma|t-s|)}{\sigma|t-s|}, \\
\nu(t) &= \frac{J_1(2\sigma t)}{\sigma t} \Theta(t).
\end{aligned} \tag{17}$$

This fully characterizes the DMFT.

### 1.3.2 Flavors of analysis.

This isn't really an essential part of the DMFT, but now that we have the effective dynamics, we can see what sort of analysis is possible. It will be useful to write the solution to the DMFT equation as:

$$x(t) = x(0)G(t) + \int_0^t dt' G(t-t') \Phi(t') \tag{18}$$

where  $G(t)$  is the Green's function of the system which satisfies:

$$\frac{dG}{dt} + \sigma^2 \int_0^t dt' \nu(t-t') G(t') = \delta(t), \quad G(0^-) = 0. \tag{19}$$

This can be solved by taking a Laplace transform:

$$\begin{aligned}
z\tilde{G}(z) + \sigma^2 \tilde{\nu}(z) \tilde{G}(z) &= 1, \quad \tilde{\nu}(z) = \frac{2}{z + \sqrt{z^2 + 4\sigma^2}} \\
\implies \tilde{G}(z) &= \frac{1}{z + \sigma^2 \tilde{\nu}(z)} \implies G(t) = \frac{J_1(2\sigma t)}{\sigma t}
\end{aligned} \tag{20}$$

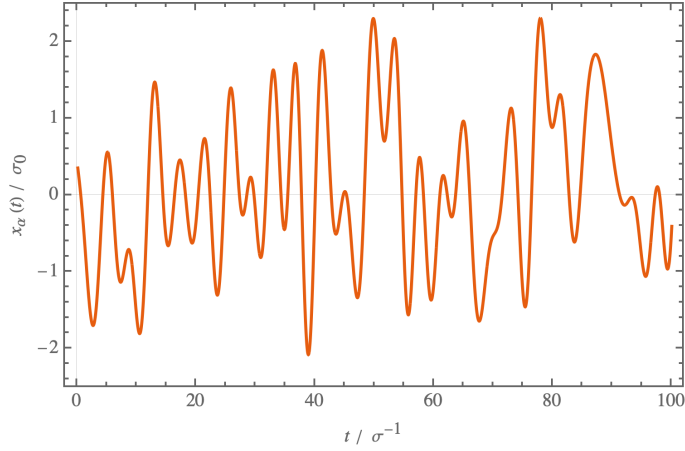


Figure 1: An example of a realization of the dynamics  $x_\alpha(t)$  for a randomly chosen constituent  $\alpha$  with  $N = 128$ .

**Covariance.** We can compute the covariance of the process  $x(t)$  as:

$$\begin{aligned} C(t, s) &= \text{cov}[x(t), x(s)]_{\text{sh}} = \langle x(t)x(s) \rangle_{\text{sh}} \\ &= \sigma_0^2 G(t)G(s) + \int_0^t dt' G(t-t') \int_0^s ds' G(s-s') \langle \Phi(t')\Phi(s') \rangle_{\text{sh}} \end{aligned} \quad (21)$$

Taking the Laplace transform in both variables, we have:

$$\begin{aligned} \tilde{C}(z_t, z_s) &= \sigma_0^2 \tilde{G}(z_t) \tilde{G}(z_s) \left[ 1 + \sigma^2 \frac{\tilde{G}(z_t) + \tilde{G}(z_s)}{z_t + z_s} \right] = \sigma_0^2 \frac{-(z_s + z_t) + \sqrt{4\sigma^2 + z_s} + \sqrt{4\sigma^2 + z_t}}{2(z_t + z_s)\sigma^2} \\ &= \sigma_0^2 \frac{G(z_t) + G(z_s)}{z_t + z_s} \\ \implies C(t, s) &= \sigma_0^2 G(|t - s|) = \sigma_0^2 \frac{J_1(2\sigma|t - s|)}{\sigma|t - s|} \end{aligned} \quad (22)$$

In the long separation limit, this becomes:

$$C(t) \sim \frac{\sigma_0^2}{\sqrt{2\pi}} \left[ \frac{\sin 2\sigma t - \cos 2\sigma t}{(\sigma t)^{3/2}} \right] \quad (23)$$

This means that  $x(t)$  has correlations that decay slowly in time and has a quasi-period of  $\pi/\sigma$ . Fig. 1 shows an example of a realization of the dynamics  $x_\alpha(t)$  for a randomly chosen constituent  $\alpha$  in which you can see the quasi-periodic oscillations.

**Power spectrum.** This finally means that the power spectrum of the process  $x(t)$  is given by:

$$S_x(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt e^{-i\omega t} C(t) = \begin{cases} \frac{\sigma_0^2}{2\pi\sigma^2} \sqrt{4\sigma^2 - \omega^2}, & \text{if } -2\sigma \leq \omega \leq 2\sigma, \\ 0, & \text{otherwise.} \end{cases} \quad (24)$$

which is exactly the distribution of the eigenvalues of the matrix  $A_{ij}$  (scaled by  $\sigma_0^2$ ).

#### 1.4 DMFT by the cavity method

Except for the linear case, you will almost never be able to do what we did above (solving the microscopic dynamics, integrating out the  $N - 1$  degrees of freedom, and then doing a statistical analysis). Instead, one takes one of two approaches: the replica method or the cavity method.

The replica method involves writing down a generating functional for the microscopic dynamics with a path integral, introducing  $n$  replicas of the system, introducing a certain replica-overlap order parameter, assuming the replica-overlap order parameter is constant over all replica indices, taking a saddle point approximation as  $N \rightarrow \infty$ , and then taking the limit  $n \rightarrow 0$  to reduce the path integral to a form that looks like a stochastic process for just one degree of freedom. The replica method is a very powerful method, but it is not very intuitive (at least to me) and intermediate steps often rely more on mathematical rather than physical arguments.

The cavity method is sometimes a more intuitive method that is based on the idea that in the thermodynamic limit, the order parameters of the system for a system of size  $N$  should be nearly the same as the order parameters of a system of size  $N + 1$ . In the cavity method, one adds a single “cavity” degree of freedom to the system and then uses linear response theory to relate the system of size  $N + 1$  to the system of size  $N$ . All this is necessary because it allows us to decorrelate certain quantities as the added constituent is definitionally independent of the system before it was added. Let’s see how this works for the linear system we have been discussing.

Just as a brief setup step, we’ll write the dynamics as:

$$\frac{dx_i}{dt} = \sum_{j=1}^N A_{ij}x_j(t) + h_i(t), \quad (25)$$

where  $h_i(t)$  is some driving force that we set to zero at the end of the calculation. It is useful to analyze perturbations to the system.

**Step 1: Adding a cavity degree of freedom.** We begin by adding a single degree of freedom,  $x_0$ , to the system. This is meant to be analogous to the randomly chosen constituent  $\alpha$  in the previous section. The dynamics of the system is now given by:

$$\frac{dx_i}{dt} = \sum_{j=1}^N A_{ij}x_j(t) + A_{i0}x_0(t) + h_i(t), \quad i = 1, \dots, N, \quad (26)$$

$$\frac{dx_0}{dt} = \sum_{i=1}^N A_{0i}x_i(t) + A_{00}x_0(t) + h_0(t). \quad (27)$$

**Step 2: Treating the cavity degree of freedom as a perturbation.** Observe that adding the cavity degree of freedom effectively perturbed the auxiliary field  $h_i$  as:

$$h_i(t) \mapsto h_i(t) + A_{i0}x_0(t) \quad (28)$$

We can treat this as a small perturbation as  $A_{i0} \sim 1/\sqrt{N}$  and  $x_0(t) \sim O(1)$ . We can then relate the dynamics of the system before and after the perturbation using a linear response approximation:

$$x_i(t) = x_{i\setminus 0}(t) + \int_0^t dt' \sum_{j=1}^N \nu_{ij}(t, t') A_{j0}x_0(t'), \quad \nu_{ij}(t, t) \equiv \left. \frac{\delta x_i(t)}{\delta h_j(t')} \right|_{h=0} \quad (29)$$

We are using the notation  $x_{i\setminus 0}(t)$  to denote the dynamics of the system without the cavity degree of freedom.

**Step 3: The cavity dynamics.** We can then substitute the linear response relation into Eq. (27) to get the dynamics of the cavity degree of freedom:

$$\frac{dx_0}{dt} = \sum_{i=1}^N A_{0i}x_{i\setminus 0}(t) + \int_0^t dt' \sum_{i,j=1}^N A_{0i}\nu_{ij}(t, t') A_{j0}x_0(t') + A_{00}x_0(t) \quad (30)$$

**Step 4: Statistical analysis.** We proceed to do a statistical analysis of the cavity degree of freedom. In this step, we investigate how each of the terms in the equation above behaves under quenched disorder. First, note that we are so far treating  $x_{i\setminus 0}(t)$  as deterministic/fixed, and it is definitionally independent of  $A_{0i}$ . This means we can approximate the first term as a Gaussian process:

$$\Phi_0(t) \equiv \sum_{i=1}^N A_{0i} x_{i\setminus 0}(t) \sim \text{Gaussian Proc.}, \quad (31)$$

which has the following statistics:

$$\begin{aligned} \langle \Phi_0(t) \rangle_{\mathcal{Q}} &= \sum_{i=1}^N \langle A_{0i} \rangle x_{i\setminus 0}(t) = 0, \\ \langle \Phi_0(t) \Phi_0(s) \rangle_{\mathcal{Q}} &= \sum_{i,j=1}^N \langle A_{0i} A_{0j} \rangle x_{i\setminus 0}(t) x_{j\setminus 0}(s) = \frac{\sigma^2}{N} \sum_{i,j=1}^N \delta_{ij} x_{i\setminus 0}(t) x_{j\setminus 0}(s) = \sigma^2 C(t, s), \end{aligned} \quad (32)$$

where we will introduce:

$$C(t, s) \equiv \frac{1}{N} \sum_{i=1}^N x_{i\setminus 0}(t) x_{i\setminus 0}(s), \quad m(t) \equiv \frac{1}{N} \sum_{i=1}^N x_{i\setminus 0}(t) \quad (33)$$

We introduce  $m(t)$  even though it is not used because it shows up in many DMFT calculations and would be relevant if we chose  $\langle A_{ij} \rangle$  to be non-zero. Next, we analyze the second term:

$$\begin{aligned} \left\langle \sum_{i,j=1}^N A_{0i} \nu_{ij}(t, t') A_{j0} \right\rangle_{\mathcal{Q}} &= \sum_{i,j=1}^N \langle A_{0i} A_{j0} \rangle \nu_{ij}(t, t') = -\frac{\sigma^2}{N} \sum_{i,j=1}^N \delta_{ij} \nu_{ij}(t, t') = -\sigma^2 \nu(t, t'), \\ \left\langle \left( \sum_{i,j=1}^N A_{0i} \nu_{ij}(t, t') A_{j0} \right)^2 \right\rangle_{\mathcal{Q}} &= \sum_{i,j,k,l=1}^N \langle A_{0i} A_{j0} A_{0k} A_{l0} \rangle \nu_{ij}(t, t') \nu_{kl}(t, t') \\ &= \frac{\sigma^4}{N^2} \sum_{i,j,k,l=1}^N (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}) \nu_{ij}(t, t') \nu_{kl}(t, t') \\ &= \sigma^4 \left( \left( \frac{1}{N} \text{Tr}[\nu(t, t')] \right)^2 + \frac{1}{N^2} \text{Tr}[\nu(t, t') \nu^T(t, t')] + \frac{1}{N^2} \text{Tr}[(\nu(t, t'))^2] \right) \\ &= (-\sigma^2 \nu(t, t'))^2 + O\left(\frac{1}{N}\right) \implies \text{var} \left[ \sum_{i,j=1}^N A_{0i} \nu_{ij}(t, t') A_{j0} \right]_{\mathcal{Q}} = O\left(\frac{1}{N}\right), \end{aligned} \quad (34)$$

where we have defined:

$$\nu(t, t') = \frac{1}{N} \sum_{i=1}^N \nu_{ii}(t, t') = \frac{1}{N} \text{Tr}[\nu(t, t')] \quad (35)$$

as the average self-response of the system. This means that the second term has negligible fluctuations in the limit  $N \rightarrow \infty$ , so we can replace it with its quenched average. The last term is  $O(N^{-1/2})$ , so we ignore it. Putting all this together, we write the dynamics of the cavity degree of freedom as:

$$\begin{aligned} \frac{dx_0}{dt} &= \Phi_0(t) - \sigma^2 \int_0^t dt' \nu(t, t') x_0(t') \\ \Phi_0(t) &\sim \text{Gaussian Proc.}, \quad \langle \Phi_0(t) \rangle_{\mathcal{Q}} = 0, \quad \langle \Phi_0(t) \Phi_0(s) \rangle_{\mathcal{Q}} = \sigma^2 C(t, s), \end{aligned} \quad (36)$$

Resampling the above stochastic dynamics is equivalent to resampling quenched disorder (the matrix  $A_{ij}$ ); the shuffling randomness is fixed.



**Step 5: Assert self-averaging to get self-consistency.** The above calculation was done essentially for just the cavity degree of freedom (i.e., fixed shuffling randomness) with fluctuating quenched disorder. We however ended up with order parameters  $m(t), C(t, s), \nu(t, s)$  that look like shuffled averages over the  $N$  constituents at fixed  $A_{ij}$ . We then assert that the system is self-averaging, meaning that we can interpret the disorder in the dynamics Eq. (36) as shuffling randomness at fixed quenched disorder. We know this is exactly true from our previous analysis of the exact solution, but typically this is an assumption that one makes. This means that every constituent  $x_i$  should have dynamics generated by the same stochastic differential equation as the cavity degree of freedom  $x_0$  (although each one will have its own realization of the stochastic process  $\Phi_i(t)$ ). The consequence is that we close the mean-field theory by writing:

$$m(t) = \langle x_0(t) \rangle_{\mathcal{Q}}, \quad C(t, s) = \langle x_0(t)x_0(s) \rangle_{\mathcal{Q}}, \quad \nu(t, s) = \left\langle \frac{\delta x_0(t)}{\delta h_0(t')} \bigg|_{h=0} \right\rangle_{\mathcal{Q}} \quad (37)$$

**Step 6: Solve the self-consistency equations (hard!).** The final step is to actually solve the self-consistency equations. This is typically the hardest part of the DMFT. First, we write a differential equation for  $\nu$ :

$$\frac{\partial}{\partial t} \nu(t, s) = \left\langle \frac{\delta}{\delta h_0(s)} \frac{dx_0(t)}{dt} \right\rangle = -\sigma^2 \int_0^t dt' \nu(t, t') \nu(t', s) + \delta(t - s) \quad (38)$$

Note that this implies that  $\nu$  is a function of  $t - s$  only, so we go to the Laplace domain:

$$z\tilde{\nu}(z) = -\sigma^2[\tilde{\nu}(z)]^2 + 1 \implies \tilde{\nu}(z) = \frac{2}{z + \sqrt{z^2 + 4\sigma^2}} \implies \nu(t) = \frac{J_1(2\sigma t)}{\sigma t} \Theta(t), \quad (39)$$

which is exactly what we found before. Yay! We can recycle our work from before to write:

$$x_0(t) = x_0(0)G(t) + \int_0^t dt' G(t - t') \Phi_0(t'), \quad G(t) = \frac{J_1(2\sigma t)}{\sigma t} \Theta(t), \quad (40)$$

Then:

$$m(t) = \langle x_0(t) \rangle = 0, \quad C(t) = \langle x_0(t)x_0(0) \rangle = \sigma_0^2 G(t) + \int_0^t dt' G(t - t') \langle \Phi_0(t')x(0) \rangle = \sigma_0^2 G(t) = \sigma_0^2 \frac{J_1(2\sigma t)}{\sigma t}, \quad (41)$$

which is exactly what we found before.

## 2 Static cavity and the perturbative resolvent method

We have seen how to use the cavity method to derive the DMFT for a linear system. Now, we will apply the cavity method to a static linear system just to provide an example of how this works in a static case. At the end, there is a very nice surprise! This is all taken from a great paper by Wenping Cui and Jason Rocks [4].

### 2.1 Setup

We'll consider the very innocent looking linear system:

$$zx_i = \sum_{j=1}^N A_{ij}x_j + h_i, \quad i = 1, \dots, N, \quad (42)$$

where  $A_{ij}$  is a symmetric matrix with i.i.d. entries with statistics:

$$\langle A_{ij} \rangle = 0, \quad \langle A_{ij}A_{kl} \rangle = \frac{\sigma^2}{N}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \quad (43)$$

As before, we'll set  $h_i = 0$  at the end of the calculation. The coefficient  $z$  is some complex number that will be important later.

## 2.2 Cavity method

Let's proceed to analyze this system using the cavity method.

**Step 1: Adding a cavity degree of freedom.** We begin by adding a single degree of freedom,  $x_0$ , to the system:

$$zx_i = \sum_{j=1}^N A_{ij}x_j + A_{i0}x_0 + h_i, \quad i = 1, \dots, N. \quad (44)$$

**Step 2: Treating the cavity degree of freedom as a perturbation.** This perturbs the auxiliary field  $h_i$  as:

$$h_i \mapsto h_i + A_{i0}x_0 \quad (45)$$

We treat this as a small perturbation:

$$x_i = x_{i\setminus 0} + \sum_{j=1}^N \nu_{ij} A_{j0}x_0, \quad (46)$$

where,

$$\nu_{ij} = \left. \frac{\partial x_i}{\partial h_j} \right|_{h=0} = \left[ \frac{1}{zI - A} \right]_{ij} \quad (47)$$

**Step 3: The cavity equation.** Substitute this into the equation for the cavity variable:

$$zx_0 = \sum_{i=1}^N A_{0i}x_{i\setminus 0} + \sum_{i,j=1}^N A_{0i}\nu_{ij}A_{j0}x_0 + A_{00}x_0 + h_0 \quad (48)$$

**Step 4: Statistical analysis.** Like before, we treat the first term as a normal random variable:

$$\Phi_0 = \sum_{i=1}^N A_{0i}x_{i\setminus 0} \sim \text{normal}, \quad (49)$$

which has the following statistics:

$$\begin{aligned} \langle \Phi_0 \rangle &= \sum_{i=1}^N \langle A_{0i} \rangle x_{i\setminus 0} = 0, \\ \langle (\Phi_0)^2 \rangle &= \sum_{i,j=1}^N \langle A_{0i}A_{0j} \rangle x_{i\setminus 0}x_{j\setminus 0} = \frac{\sigma^2}{N} \sum_{i,j=1}^N \delta_{ij}x_{i\setminus 0}x_{j\setminus 0} = \sigma^2 q, \end{aligned} \quad (50)$$

where  $q = \frac{1}{N} \sum_{i=1}^N x_{i\setminus 0}^2$ . The second term has the following statistics:

$$\begin{aligned} \left\langle \sum_{i,j=1}^N A_{0i}\nu_{ij}A_{j0} \right\rangle &= \sum_{i,j=1}^N \langle A_{0i}A_{j0} \rangle \nu_{ij} = \sigma^2 \bar{\nu}, \\ \left\langle \left( \sum_{i,j=1}^N A_{0i}\nu_{ij}A_{j0} \right)^2 \right\rangle &= \sum_{i,j,k,l=1}^N \langle A_{0i}A_{j0}A_{0k}A_{l0} \rangle \nu_{ij}\nu_{kl} = \frac{\sigma^4}{N^2} \sum_{i,j,k,l=1}^N (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \nu_{ij}\nu_{kl} \\ &= \sigma^4 \left( \left( \frac{1}{N} \text{Tr}[\nu] \right)^2 + \frac{1}{N^2} \text{Tr}[\nu \nu^T] + \frac{1}{N^2} \text{Tr}[\nu^2] \right) \\ &= (\sigma^2 \bar{\nu})^2 + O\left(\frac{1}{N}\right), \end{aligned} \quad (51)$$

where we have defined:

$$\bar{\nu} = \frac{1}{N} \sum_{i=1}^N \nu_{ii} = \frac{1}{N} \text{Tr}[\nu] = \frac{1}{N} \sum_{i,j=1}^N \left[ \frac{1}{zI - A} \right]_{ii}. \quad (52)$$

Putting it all together:

$$\begin{aligned} zx_0 &= \Phi_0 + \sigma^2 \bar{\nu} x_0 + h_0 \\ \Phi_0 &\sim \mathcal{N}(0, \sigma^2 q). \end{aligned} \quad (53)$$

**Step 5: Assert self-averaging to get self-consistency.** We close the mean-field theory by writing:

$$\begin{aligned} q &= \langle x_0^2 \rangle = \left\langle \left[ \frac{\Phi_0}{z - \sigma^2 \bar{\nu}} \right]^2 \right\rangle = \frac{q^2 \sigma^4}{(z - \nu \sigma^2)^2}, \\ \bar{\nu} &= \left\langle \frac{\partial x_0}{\partial h_0} \right\rangle_{h=0} = \left\langle \frac{\partial}{\partial h_0} \frac{\Phi_0 + h_0}{z - \sigma^2 \bar{\nu}} \right\rangle_{h=0} = \frac{1}{z - \sigma^2 \bar{\nu}} \end{aligned} \quad (54)$$

Solving the self-consistency equations gives:

$$\bar{\nu}(z) = \frac{1}{N} \text{Tr} \left[ \frac{1}{zI - A} \right] = \frac{z \pm \sqrt{z^2 - 4\sigma^2}}{2\sigma^2}, \quad q = \left( \frac{z \mp \sqrt{z^2 - 4\sigma^2}}{2\sigma^2} \right)^2 \quad (55)$$

The branch we take depends on the value of  $z$ , but it is typically the case that we take the branch cut so that  $\text{Im}[\bar{\nu}(x - i0^+)] \geq 0$  with  $x \in \mathbb{R}$ .

### 2.3 The spectrum of a random matrix

First, remember the Sokhotski—Plemelj identity:

$$\lim_{\epsilon \rightarrow 0^+} \frac{1}{x \pm i\epsilon} = \mp i\pi \delta(x) + \mathcal{P} \frac{1}{x} \quad (56)$$

which is interpreted in the distributional sense.  $\mathcal{P}$  denotes the Cauchy principal value. It follows that:

$$\begin{aligned} \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \text{Im} \left[ \frac{1}{N} \text{Tr} \left[ \frac{1}{(x - i\epsilon)I - A} \right] \right] &= \frac{1}{\pi} \text{Im} \left[ \lim_{\epsilon \rightarrow 0^+} \frac{1}{N} \sum_{i=1}^N \frac{1}{x - \lambda_i - i\epsilon} \right] \\ &= \frac{1}{\pi} \text{Im} \left[ \frac{1}{N} \sum_{i=1}^N i\pi \delta(x - \lambda_i) \right] = \rho_A(x). \end{aligned} \quad (57)$$

This means that the spectral density of the random matrix  $A$  is given by:

$$\rho_A(x) = \lim_{\epsilon \rightarrow 0^+} \frac{1}{\pi} \text{Im} [\bar{\nu}(x - i\epsilon)] = \begin{cases} \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - x^2}, & \text{if } -2\sigma \leq x \leq 2\sigma, \\ 0, & \text{otherwise.} \end{cases} \quad (58)$$

We have derived the Wigner semicircle law! This can be done for many other random matrix ensembles including non-symmetric matrices (but the identity to get  $\rho_A(x + iy)$  is slightly different). See Ref. [4] for a more complete description of this technique and several examples.

## 3 What's next?

The next pedagogical step is to apply the cavity method to a nonlinear system. In these sorts of nonlinear systems, there can be certain sorts of phase transitions that are describable using the cavity method. For an example of this sort of analysis, see Ref. [5, 8, 9, 10]. It is also useful to look at some spin glass literature to see where this all comes from [1].

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